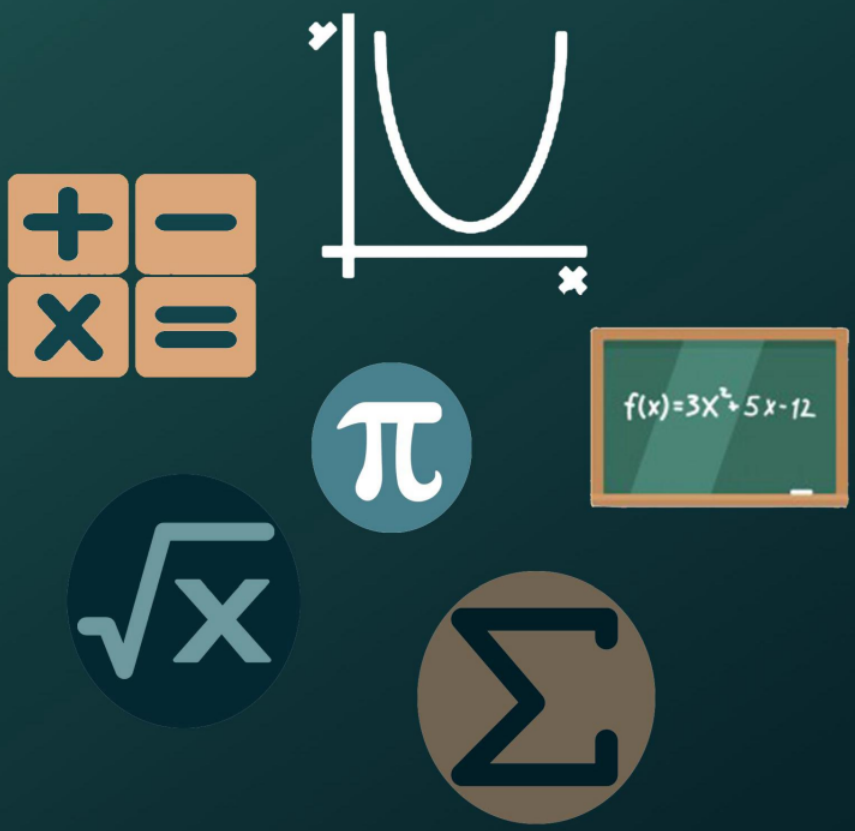
MATHEMATICS

Binomial Theorem



SHEET

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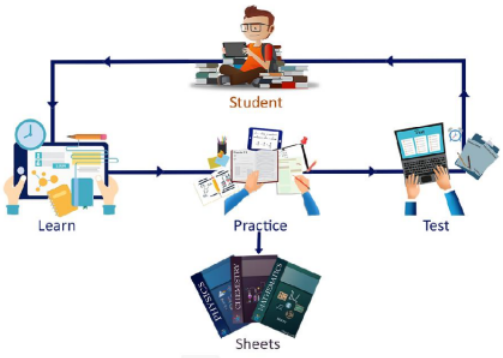
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Binomial Theorem				
2	Expansion of Binomial Theorem	1-4			
3	m^{th}-term from the End	5-6	1-2, 17		
4	Coefficient of x to the power m in the Expansion	7-11	20	1-5	1
5	Middle Term in the Expansion	12-14	3,4		
6	Number of Terms in the Expansion	15,16	5,21	6-8	19-21
7	Numerically Greatest Term	17	9,18	9	10,11
8	Properties of Binomial Coefficient -Type 1	18,19		10-13	
9	Properties of Binomial Coefficient -Type 2	20-23		14	22,25
10	Properties of Binomial Coefficient -Type 3	24-26			26
11	Properties of Binomial Coefficient -Type 4	27-30	15,22-24	15	17,27
12	Sum of Coefficients in an Expansion	31,32	9,10	16-18	
13	Applications of Binomial Theorem	33-36	25	19-23	2,3
14	Multinomial Theorem	37-39	6,11-14, 26		4-7,12-16, 18
15	Binomial Theorem for Negative & Fractional Index		7		8
16	Approximation				
17	After Completing Chapter	40	8,16,19	24-32	9,23,24, 28-30

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Expansion of Binomial Theorem

Q1. If $(1+ax)^n = 1+8x+24x^2+\ldots\ldots$, then the values of 'a' and 'n' are equal to -

(A) 1, 2 (B) 3, 6 (C) 2, 3 (D) 2, 4

Q2. If x^p occurs in the expansion of $\left(x^2+\frac{1}{x}\right)^{2n}$, its coefficient is

(A) ${}^{2n}C_{\frac{4n-p}{3}}$ (B) ${}^{2n}C_{\frac{2n-p}{3}}$

(C) ${}^{2n}C_{4n-\frac{p}{3}}$ (D) None of these

Q3. If the coefficients of second, third and fourth terms in the expansion of $(1+x)^{2n}$ are in A. P., then-

(A) $2n^2-9n+7=0$ (B) $2n^2+5n+7=0$

(C) $n^2-9n+7=0$ (D) None of these

Q4. The 9th term in the expansion of $\left(\frac{x}{a}-\frac{3a}{x^2}\right)^{12}$ is-

(A) ${}^{12}C_4x^{12}a^4\,3^8$ (B) ${}^{12}C_4x^{-12}a^4\,3^8$

(C) ${}^{12}C_4x^{-12}a^4$ (D) None of these

m^{th} term from the End

Q5. If in the expansion of $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the 7th term from the beginning to the seventh term from the end is equal to $\frac{1}{6}$, then n can be equal to -

(A) 3 (B) 6

(C) 9 (D) None of these

Q6. The 11th term from the end in the expansion of $\left(2x-\frac{1}{x^2}\right)^{25}$ is -

(A) ${}^{25}C_{15}\,2^9\,x^{-20}$ (B) ${}^{25}C_{15}\,2^{10}\,x^{-20}$

(C) $-{}^{25}C_{15}\,2^{10}\,x^{-20}$ (D) None of these

Coefficient of x to the power m in the Expansion

Q7. The term independent of x in $\left[\sqrt{\left(\frac{x}{3}\right)}+\sqrt{\left(\frac{3}{2x^2}\right)}\right]^{10}$ is -

(A) 1 (B) ${}^{10}C_1$

(C) $\frac{5}{12}$ (D) None of these

Q8. The coefficient of $\frac{1}{y^2}$ in $\left(y+\frac{c^3}{y^2}\right)^{10}$ is -

(A) $187\,c^{12}$ (B) $\frac{210}{c^{12}}$

(C) $\frac{c^{12}}{210}$ (D) $210\,c^{12}$

Q9. If the rth term in the expansion of $\left(\frac{x}{3}-\frac{2}{x^2}\right)^{10}$ contains x^4, then r is equal to -

(A) 2 (B) 3

(C) 4 (D) 5

Q10. The coefficient of x^7 in the expansion of $(1-x^4)(1+x)^9$ is -

(A) –48 (B) 48

(C) 9 (D) 27

Q11. The coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^{11}$ is -

(A) 990 (B) 605

(C) 810 (D) None of these

Middle Term in the Expansion

Q12. The largest coefficient in the expansion of $(1+x)^{24}$ is -

(A) ${}^{24}C_{13}$ (B) ${}^{24}C_{11}$

(C) ${}^{24}C_{24}$ (D) ${}^{24}C_{12}$

Q13. The middle terms in the expansion of $\left(\frac{3}{x^2}-\frac{x^3}{6}\right)^9$ are -

(A) $\frac{189}{8}x^2, -\frac{21}{16}x^7$

(B) $\frac{189}{8}x^3, -\frac{21}{16}x^6$

(C) $\frac{189}{21}x^2, -\frac{21}{16}x^5$

(D) None of these

Q14. If the middle term in the expansion of $\left(x^2+\frac{1}{x}\right)^n$ is $924x^6$, then n =

(A) 10 (B) 12

(C) 14 (D) None of these

Number of Terms in the Expansion

Q15. The expression

$$\left[x+\left(x^3-1\right)^{1/2}\right]^5+\left[x-\left(x^3-1\right)^{1/2}\right]^5$$

is a polynomial of degree.

(A) 5 (B) 6

(C) 7 (D) 8

Q16. The number of non-zero terms in the expansion of $\left(1+3\sqrt{2}x\right)^9+\left(1-3\sqrt{2}x\right)^9$ is -

(A) 9 (B) 0 (C) 5 (D) 10

Numerically Greatest Term

Q17. The greatest term (numerically) in the expansion of $(3-5x)^{11}$ when $x=\frac{1}{5}$ is -

(A) 55×3^9 (B) 46×3^9

(C) 55×3^6 (D) None of these

Properties of Binomial Cofficient - Type 1

Q18. ${}^nC_n+{}^{n+1}C_n+{}^{n+2}C_n+....+{}^{n+k}C_n=$

(A) ${}^{n+k-1}C_{n+1}$ (B) ${}^{n+k}C_{n+1}$

(C) ${}^{n+k+1}C_{n+1}$ (D) None of these

Q19. If $(1+x)^n=C_0+C_1x+C_2x^2+...C_nx^n$, then the value of

$(C_0+C_1)(C_1+C_2).....(C_{n-1}+C_n)$

$=k.\,C_1\,C_2\,C_3...C_n$

where k =

(A) $\frac{(n+1)^n}{n!}$ (B) $\frac{n^n}{(n-1)!}$

(C) $\frac{(n+1)^n}{(n-1)!}$ (D) none of these

Properties of Binomial Cofficient - Type 2

Q20. $\frac{C_1}{C_0}+2\frac{C_2}{C_1}+3\frac{C_3}{C_2}+....n\frac{C_n}{C_{n-1}}=$

(A) $\frac{n(n-1)}{2}$ (B) $\frac{n(n+1)}{2}$

(C) $\frac{(n+1)(n+2)}{2}$ (D) None of these

Q21. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $\sum_{k=1}^{n} k^2.C_k =$

(A) $n(n-1)2^{n-1}$ (B) $n(n+1)2^{n-2}$

(C) $(n+1)(n+2)2^{n-1}$ (D) None of these

Q22. $1.{}^nC_1 + 2.{}^nC_2 + 3.{}^nC_3 + ... + n.{}^nC_n$ is equal to-

(A) $\frac{n(n+1)}{4}.2^n$ (B) $2^{n+1} - 3$

(C) $n.2^{n-1}$ (D) None of these

Q23. $aC_0 + (a + b)C_1 + (a + 2b)C_2 + ...+ (a + nb)C_n$ is equal to-

(A) $(2a + nb)2^n$ (B) $(2a + nb)2^{n-1}$

(C) $(na + 2b)2^n$ (D) $(na + 2b)2^{n-1}$

Properties of Binomial Cofficient - Type 3

Q24. If $C_0, C_1, C_2, ..., C_n$ are the coefficients of the expansion of $(1+x)^n$, then the value of $\sum_{0}^{n} \frac{C_k}{k+1}$ is-

(A) 0 (B) $\frac{2^n - 1}{n}$

(C) $\frac{2^{n+1} - 1}{n+1}$ (D) None of these

Q25. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $2C_0 + 2^2.\frac{C_1}{2} + 2^3.\frac{C_2}{3} + ... + 2^{n+1}.\frac{C_n}{n+1} =$

(A) $\frac{3^{n+1} - 1}{n+1}$ (B) $\frac{3^n - 1}{n}$

(C) $\frac{3^{n+2} - 1}{n+2}$ (D) None of these

Q26. The value of $\frac{C_1}{2} + \frac{C_3}{4} + \frac{C_5}{6} + ...$ is equal to -

(A) $\frac{2^n + 1}{n+1}$ (B) $\frac{2^n - 1}{n+1}$

(C) $\frac{2^n + 1}{n-1}$ (D) $\frac{2^n}{n+1}$

Properties of Binomial Cofficient - Type 4

Q27. $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $C_0C_2 + C_1C_3 + C_2C_4 + ... + C_{n-2}C_n =$

(A) $\frac{(2n)!}{(n!)^2}$ (B) $\frac{(2n)!}{(n-1)!(n+1)!}$

(C) $\frac{(2n)!}{(n-2)!(n+2)!}$ (D) None of these

Q28. ${}^mC_r + {}^mC_{r-1}.{}^nC_1 + {}^mC_{r-2}.{}^nC_2 + ...$

$+ {}^mC_1.{}^nC_{r-1} + {}^nC_r =$

(A) ${}^{m+n}C_{r-1}$

(B) ${}^{m+n}C_r$

(C) ${}^{m+n}C_{r+1}$

(D) None of these

Q29. The value of the sum of the series ${}^{14}C_0.{}^{15}C_1 + {}^{14}C_1.{}^{15}C_2 + {}^{14}C_2.{}^{15}C_3 + ...$

$+ {}^{14}C_{14}.{}^{15}C_{15}$

is -

(A) ${}^{29}C_{12}$ (B) ${}^{29}C_{10}$

(C) ${}^{29}C_{14}$ (D) ${}^{29}C_{16}$

Q30. If $(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, then $C_0^2 + C_1^2 + C_2^2 + ... + C_n^2 =$

(A) $\frac{(2n)!}{(n!)^2}$ (B) $\frac{(2n)!}{(n-1)!(n+1)!}$

(C) $\frac{(2n)!}{(n-2)!(n+2)!}$ (D) None of these

Sum of Coefficients in an Expansion

Q31. The sum of the coefficients in the expansion of $(1+5x-7x^3)^{3165}$ is -

(A) 1 (B) 2^{3165}

(C) 2^{3164} (D) -1

Q32. If the sum of the coefficients in the expanssion of $(\alpha^2x^2 - 2\alpha x + 1)^{51}$ vanishes, then α is equal to-

(A) -2 (B) 2

(C) 1 (D) -1

Applications of Binomial Theorem

Q33. The sum of rational terms in the expansion of $(\sqrt{2} + 3^{1/5})^{10}$ is -

(A) 31 (B) 41

(C) 51 (D) None of these

Q34. The fractional part of $\frac{2^{4n}}{15}$ is -

(A) $\frac{1}{15}$ (B) $\frac{2}{15}$

(C) $\frac{4}{15}$ (D) None of these

Q35. For all $n \in N$, $2^{4n} - 15n - 1$ is divisible by :

(A) 225 (B) 125

(C) 325 (D) None of these

Q36. If 7^{103} is divided by 25, then the remainder is -

(A) 20 (B) 16

(C) 18 (D) 15

Multinomial Theorem

Q37. The coefficient of $x^3\ y^4\ z^2$ in the expansion of $(2x - 3y + 4z)^9$ is -

(A) 12063600

(B) 13063680

(C) 11063600

(D) 680

Q38. If $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, then $a_0 + a_3 + a_6 + ... =$ ($n \in N$, n is multiple of 3).

(A) 3^{n+1} (B) 3^n

(C) 3^{n-1} (D) None of these

Q39. If $(1-x+x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, then $a_0 + a_2 + a_4 + ... + a_{2n}$ is equal to -

(A) $3n^2 + \frac{1}{2}$ (B) $\frac{1-3^n}{2}$

(C) $\frac{3^n - 1}{2}$ (D) $\frac{3^n + 1}{2}$

After Complete Chapter

Q40. In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of 5th and 6th terms is zero, then $\frac{a}{b}$ equals:

(A) $\frac{6}{n-5}$ (B) $\frac{n-5}{6}$

(C) $\frac{n-4}{5}$ (D) $\frac{5}{n-4}$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If the coefficients of three consecutive terms in the expansion of $(1 + x)^n$ are in the ratio of 1 : 7 : 42, then n is divisible by -
(A) 9 (B) 5 (C) 3 (D) 11

Q2. In the expansion of $\left(x^{2/3} - \frac{1}{\sqrt{x}}\right)^{30}$, a term containing the power x^{13}.
(A) does not exist
(B) exists and the co-efficient is divisible by 29
(C) exists and the co-efficient is divisible by 63
(D) exists and the co-efficient is divisible by 65

Q3. The middle term in the expansion of $\left(x^2 + \frac{1}{x^2} + 2\right)^n$ is -
(A) $\frac{n!}{\left[\left(\frac{n}{2}\right)!\right]^2}$ (B) $\frac{2n!}{\left[\left(\frac{n}{2}\right)!\right]^2}$
(C) $\frac{1.3.5...(2n-1)}{n!}2^n$ (D) $\frac{(2n)!}{(n!)^2}$

Q4. The co-efficient of the middle term in the expansion of $(1+x)^{2n}$ is -
(A) $\frac{1.3.5.7.....(2n-1)}{n!}\,2^n$
(B) ${}^{2n}C_n$
(C) $\frac{(n+1)\,(n+2)\,(n+3)\,....\,(2n-1)\,(2n)}{1.2.3.....\,(n-1)\,n}$
(D) $\frac{2.6.10.14.....\,(4n-6)\,(4n-2)}{1.2.3.4.....(n-1)\,.\,n}$

Q5. In the expansion of $\left(\sqrt[3]{4} + \frac{1}{\sqrt[4]{6}}\right)^{20}$ -
(A) the number of irrational terms = 19
(B) middle term is irrational
(C) the number of rational terms = 2
(D) 9th term is rational

Q6. In the expansion of $(a + b + c)^{10}$
(A) total number of terms is 66
(B) coefficient of $a^8\,bc$ is 90
(C) coefficient of $a^4\,b^5\,c^3$ is 0
(D) None of these

Q7. Number of terms free from radical sign in the expansion of $(1 + 3^{1/3} + 7^{1/7})^{10}$ is -
(A) 4 (B) 5 (C) 6 (D) 8

Q8. The coefficient of $x^r\ (0 \le r \le n-1)$ in the expression :
$(x+2)^{n-1} + (x+2)^{n-2}\,.\,(x+1) + (x+2)^{n-3}\,.\,(x+1)^2 + ... + (x+1)^{n-1}$ is -
(A) ${}^nC_r(2^r-1)$ (B) ${}^nC_r(2^{n-r}-1)$
(C) ${}^nC_r(2^r+1)$ (D) ${}^nC_r(2^{n-r}+1)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four options corresponds to the correct answer. For each question, choose the options corresponding to the correct answer.*

Paragraph # 1 (Ques. 9 to 10)

In the expansion of $(3^{-x/4} + 3^{5x/4})^n$, the sum of the binomial coefficient is 64 and the term with greatest binomial coefficient exceeds the third by (n–1) then-

Q9. The value of n is divisible by -
(A) 2 (B) 3
(C) 6 (D) 4

Q10. If Tp be the term having greatest binomial coefficient then the value of p is-

(A) 5

(B) odd number

(C) even number

(D) 4

Paragraph # 2 (Ques.11 to 14)

If n is positive integer and if

$$(1 + 4x + 4x^2)^n = \sum_{r=0}^{2n} a_r x^r ,$$

where a_i's are (i = 0, 1, 2, 3, ... , 2n) real numbers. On the basis of above information, answer the following questions :

Q11. The value of $2\sum_{r=0}^{n} a_{2r}$ is -

(A) $9^n - 1$ (B) $9^n + 1$

(C) $9^n - 2$ (D) $9^n + 2$

Q12. The value of $2\sum_{r=1}^{n} a_{2r-1}$ is -

(A) $9^n - 1$ (B) $9^n + 1$

(C) $9^n - 2$ (D) $9^n + 2$

Q13. The value of a_{2n-1} is -

(A) 2^{2n} (B) $(n-1).2^{2n}$

(C) $n.2^{2n}$ (D) $(n+1).2^{2n}$

Q14. The value of a_2 is -

(A) 8n (B) $8n^2 - 4$

(C) $8n^2 - 4n$ (D) 8n – 4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15.

Column-I	**Column-II**
(A) $(2n+1)(2n+3)(2n+5) \ldots\ldots (4n-1)$ is equal to	(P) $\frac{(n+1)^n}{n!}$
(B) $\frac{C_1}{C_0} + \frac{2 . C_2}{C_1} + \frac{3 . C_3}{C_2} + \ldots\ldots + \frac{n . C_n}{C_{n-1}}$ is equal to here C_r stand for ${}^nC_r$.	(Q) $n . 2^n . (2^n - 1)$
(C) If $(C_0+C_1)(C_1+C_2)(C_2+C_3) \ldots\ldots (C_{n-1}+C_n) = m .$ $C_1C_2C_3 \ldots . C_n$, then m is equal to	(R) $\frac{(4n)!\ n!}{2^n . (2n)!\ (2n)!}$
(D) If C_r are the binomial co–efficients in the expansion of $(1 + x)^n$, the value of $\sum_{i=1}^{n}\sum_{j=1}^{n}(i+j)C_iC_j$ is	(S) $\frac{n(n+1)}{2}$

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q16. Match list-I with list-II and select the correct answer using the code given below the list.

List-I	List-II
(P) The term independent of x in $\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is T_k, then k is equal to-	(i) 14
(Q) The coefficients of 9th, 10th and 11th terms in the expansion $(1+x)^n$ are in AP, then n =	(ii) 24
(R) The first integral term in the expansion of $\left(\sqrt{3}+\sqrt[3]{2}\right)^9$ is T_r, then value of r	(iii) 7
(S) In the expansion of $\left(\frac{a}{x}+bx\right)^{12}$, the coefficient of x^{-10} will be q at a = 1, b = 2 then q =	(iv) 4

Codes:

	(P)	(Q)	(R)	(S)		(P)	(Q)	(R)	(S)
(A)	(iii)	(ii)	(iv)	(i)	(B)	(iv)	(i)	(iii)	(ii)
(C)	(iii)	(i)	(iv)	(ii)	(D)	(ii)	(iv)	(i)	(iii)

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q17. If the coefficients of $(2r+4)^{th}$, $(r-2)^{th}$ terms in the expansion of $(1+x)^{18}$ are equal, find r.

Q18. Find the index n of the binomial $\left(\frac{x}{5}+\frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient ($n \in N$).

Q19. If $^{40}C_1 \cdot x(1-x)^{39} + 2 \cdot {}^{40}C_2\, x^2(1-x)^{38} + 3\, {}^{40}C_3\, x^3(1-x)^{37} + \ldots + 40 \cdot {}^{40}C_{40}\, x^{40} = ax + b$, then find the value of $(a+b)$.

SUBJECTIVE TYPE

Q20. Find the term independent of x in the expansion of :

(a) $\left[\sqrt{\frac{x}{3}}+\frac{\sqrt{3}}{2x^2}\right]^{10}$ (b) $\left[\frac{1}{2}x^{1/3}+x^{-1/5}\right]^8$

Q21. Prove that the ratio of the coefficient of x^{10} in $(1-x^2)^{10}$ & the term independent of x in $\left(x-\frac{2}{x}\right)^{10}$ is 1 : 32.

Q22. Prove that : $\sum_{r=0}^{n-2}\binom{n-1}{r}\binom{n}{r+2}=\binom{2n-1}{n-2}$

Q23. Prove that :

$$C_0C_r + C_1C_{r+1} + C_2C_{r+2} + \ldots\ldots + C_{n-r}C_n = \frac{2n!}{(n-r)!(n+r)!}$$

Q24. Evaluate :

$$2^{15}\binom{30}{0}\binom{30}{15} - 2^{14}\binom{30}{1}\binom{29}{14} + 2^{13}\binom{30}{2}\binom{28}{13} \ldots\ldots - \binom{30}{15}\binom{15}{0}$$

Q25. Which is larger : $(99^{50}+100^{50})$ or $(101)^{50}$.

Q26. Find the coefficient of x^4 in the expansion of $(2-x+3x^2)^6$.

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If the coefficents of x^2 and x^3 in the expansion of $(3 + kx)^9$ are equal then the value of k is

(A) $\frac{7}{9}$ (B) $-\frac{9}{8}$ (C) $\frac{9}{7}$ (D) $\frac{7}{9}$

Q2. If in the expansion of $(1+x)^m(1-x)^n$ the coefficients of x and x^2 are 3 and –6 respectively, then m is -

(A) 6 (B) 9 (C) 12 (D) 24

Q3. The coefficient of x^{20} in the expansion of $(1 + x^2)^{40} \cdot \left(x^2 + 2 + \frac{1}{x^2}\right)^{-5}$ is -

(A) ${}^{30}C_{10}$ (B) ${}^{30}C_{25}$

(C) 1 (D) 0

Q4. If $a^3 + b^6 = 2$, then the maximum value of the term independent of x in the expansion of $(ax^{1/3} + bx^{-1/6})^9$ is $(a > 0, b > 0)$.

(A) 42 (B) 68 (C) 84 (D) 148

Q5. If the coefficient of x^7 in $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation-

(A) ab = 1 (B) $\frac{a}{b} = 1$

(C) a + b = 1 (D) a – b = 1

Q6. The number of integral terms in the expansion of $\left(\sqrt{3} + \sqrt[8]{5}\right)^{256}$ is -

(A) 32 (B) 33 (C) 34 (D) 35

Q7. The number of terms in the expansion of $\left(1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5\right)^{20}$ is -

(A) 100 (B) 101

(C) 120 (D) None of these

Q8. The number of terms in $(1+x)^{101}\left(1+x^2-x\right)^{100}$ is -

(A) 302 (B) 301

(C) 202 (D) 101

Q9. If the largest interval to which x belongs so that the greatest term in $(1+x)^{2n}$ has the greatest coefficient is $\left(\frac{10}{11}, \frac{11}{10}\right)$, then n =

(A) 9 (B) 10

(C) 11 (D) None of these

Q10. The sum of the series $\sum_{r=0}^{10} {}^{20}C_r$ is -

(A) $2^{19} - \frac{1}{2} \cdot {}^{20}C_{10}$ (B) $2^{19} + \frac{1}{2} \cdot {}^{20}C_{10}$

(C) 2^{19} (D) 2^{20}

Q11. In the expansion of $(1 + x)^{50}$, the sum of the coefficient of odd power of x is

(A) 0 (B) 2^{50}

(C) 2^{49} (D) 2^{51}

Q12. The value of ${}^{35}C_8 + \sum_{r=1}^{7} {}^{42-r}C_7 + \sum_{r=1}^{5} {}^{47-r}C_{40-r}$ is equal to

(A) ${}^{47}C_8$ (B) ${}^{47}C_9$

(C) ${}^{47}C_7$ (D) None of these

Q13. The value of the expression ${}^{n+1}C_2 + 2\,[{}^2C_2 + {}^3C_2 + {}^4C_2 + \ldots\ldots + {}^nC_2]$ is -

(A) Σn (B) Σn^2

(C) Σn^3 (D) $\frac{(n+1)}{2}$

Q14. If n is a positive integer greater than 1, then $a - {}^nC_1(a-1) + {}^nC_2(a-2) - .... + (-1)^n(a-n)\,{}^nC_n$ is equal to

(A) n (B) a

(C) 0 (D) None of these

Q15. If $A = {}^{2n}C_0 \cdot {}^{2n}C_1 + {}^{2n}C_1\,{}^{2n-1}C_1 + {}^{2n}C_2\,{}^{2n-2}C_1 + ...$ then A is -

(A) 0 (B) 2^n (C) $n2^{2n}$ (D) 1

Q16. If the sum of the coefficients in the expansion of $(\alpha x^2 - 2x + 1)^{35}$ is equal to the sum of the coefficients in the expansion of $(x - \alpha y)^{35}$, then α is equal to -

(A) 3 (B) 2 (C) 1 (D) –2

Q17. If the sum of the coefficients in the expansion of $(1 + 2x)^m$ and $(2 + x)^n$ are respectively 6561 and 243, then the position of the point (m, n) with respect to the circle: $x^2 + y^2 - 4x - 6y - 32 = 0$.

(A) is inside the circle

(B) is outside the circle

(C) is on the circle

(D) cannot be fixed

Q18. The sum of the last ten coefficients in the expansion of $(1 + x)^{19}$, when expanded in ascending powers of x is -

(A) 2^{18} (B) 2^{19}

(C) $2^{18} - {}^{19}C_{10}$ (D) None of these

Q19. The greatest integer which divides the number $101^{100} - 1$ of the following

(A) 100 (B) 1000

(C) 10000 (D) 100000

Q20. Let $R = \left(5\sqrt{5} + 11\right)^{2n+1}$ and $f = R - [R]$ where[ ] denotes the greatest integer function, then $R.f =$

(A) 2^{2n+1} (B) 2^{4n+1}

(C) 4^{2n+1} (D) None of these

Q21. The remainder when 3^{37} is divided by 80 is -

(A) 78 (B) 3

(C) 2 (D) 35

Q22. If 5^{40} is divided by 11, then remainder is α and when 2^{2003} is divided by 17, then remainder is β, then the value of $\beta - \alpha$ is

(A) 3 (B) 5 (C) 7 (D) 5

Q23. If $\left(5 + 2\sqrt{6}\right)^n = I + f;\ n, I \in N;$ and $0 \le f < 1$, then I equals :

(A) $\frac{1}{f} - f$ (B) $\frac{1}{1+f} - f$

(C) $\frac{1}{1+f} + f$ (D) $\frac{1}{1-f} - f$

Q24. The coefficient of x^{100} in the expansion of $\sum_{j=0}^{200}(1+x)^j$ is -

(A) $\binom{200}{100}$ (B) $\binom{201}{102}$

(C) $\binom{200}{101}$ (D) $\binom{201}{100}$

Q25. If $s_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ and $t_n = \sum_{r=0}^{n} \frac{r}{{}^nC_r}$ then $\frac{t_n}{s_n}$ is-

(A) $n - 1$ (B) $\frac{1}{2}n - 1$

(C) $\frac{1}{2}n$ (D) $\frac{2n-1}{2}$

Q26. The expression $({}^{10}C_0)^2 - ({}^{10}C_1)^2 + ... - ({}^{10}C_9)^2 + ({}^{10}C_{10})^2$ equals -

(A) ${}^{10}C_5$ (B) $-{}^{10}C_5$

(C) $({}^{10}C_5)^2$ (D) $(10!)^2$

Q27. The sum of the series
$^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots\ldots - \ldots\ldots + {}^{20}C_{10}$ is -

(A) $\frac{1}{2}{}^{20}C_{10}$ (B) 0

(C) $-{}^{20}C_{10}$ (D) $^{20}C_{10}$

Q28. For $2 \le r \le n$, $\binom{n}{r}+2\binom{n}{r-1}+\binom{n}{r-2}=$

(A) $\binom{n+1}{r-1}$ (B) $2\binom{n+1}{r+1}$

(C) $2\binom{n+2}{r}$ (D) $\binom{n+2}{r}$

Q29. Coefficient of t^{24} in the expansion of $(1 + t^2)^{12} (1 + t^{12}) (1 + t^{24})$ is -

(A) $^{12}C_6 + 2$ (B) $^{12}C_6 + 1$

(C) $^{12}C_6$ (D) none

Q30. If n, r $\in$ N and $^{n-1}C_r = (k^2 - 3)\,(^nC_{r+1})$, then k lies in the interval -

(A) $\left[-\sqrt{3}, \sqrt{3}\right]$ (B) $(2, \infty)$

(C) $\left[-\sqrt{3}, \infty\right)$ (D) $\left(\sqrt{3}, 2\right]$

Q31. The value of

$$\binom{30}{0}\binom{30}{10}-\binom{30}{1}\binom{30}{11}+\binom{30}{2}\binom{30}{12}\ldots\ldots$$

$$+\binom{30}{20}\binom{30}{30}, \text{ is where } \binom{n}{r} = {}^nC_r$$

(A) $^{30}C_{10}$ (B) $^{60}C_{20}$

(C) $^{31}C_{11}$ or $^{31}C_{10}$ (D) $^{30}C_{11}$

Q32. The coefficient of x^{17} in the expansion of $(x - 1)(x - 2)(x - 3) \ldots (x - 18)$ is -

(A) $\frac{171}{2}$ (B) 342

(C) –171 (D) 684

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. In the expansion of $\left(x^3 + 3.2^{-\log_{\sqrt{2}}\sqrt{x^3}}\right)^{11}$ -

(A) there appears a term containing x^2

(B) there does not appear a term containing x^2

(C) there appears a term with the power x^{-3}

(D) the ratio of the co-efficient of x^3 to that of x^{-3} is $\frac{1}{3}$

Q2. If $\left(8+3\sqrt{7}\right)^n = P + F$, where P is an integer and F is a proper fraction, then -

(A) P is an odd integer

(B) P is an even integer

(C) F. (P + F) = 1

(D) (1– F)(P + F) = 1

Q3. The last digit of $3^{3^{4n}}+1, n \in N$, is -

(A) $^4C_3$ (B) $^8C_7$ (C) 8 (D) 4

Q4. If $(1 + 2x + 3x^2)^{10} = a_0 + a_1x + a_2x^2 + \ldots. + a_{20}x^{20}$, then

(A) $a_1 = 20$ (B) $a_2 = 210$

(C) $a_4 = 8085$ (D) $a_{20} = 2^2.3^7.7$

Q5. If $(1+x+x^2)^{25} = a_0 + a_1x + a_2x^2 + ... + a_{50}.x^{50}$ then $a_0 + a_2 + a_4 + ... + a_{50}$ is -

(A) even

(B) odd & of the form 3n

(C) odd & of the form $(3n-1)$

(D) odd & of the form $(3n+1)$

Q6. Let $(1 + x^2)^2 (1 + x)^n = A_0 + A_1 x + A_2 x^2 + ...$ If A_0, A_1, A_2 are in A.P. then the value of n can be equal to-

(A) 2 (B) 3 (C) 5 (D) 7

Q7. If $(1 + x + x^2 + x^3)^{100} = a_0 + a_1x + a_2x^2 + ... + a_{300}x^{300}$, then -

(A) $a_0 + a_1 + a_2 + a_3 + ...+ a_{300}$ is divisible by 1024

(B) $a_0 + a_2 + a_4 + ...+ a_{300} = a_1 + a_3 + ... + a_{299}$

(C) coefficients equidistant from beginning and end are equal

(D) $a_1 = 100$

Q8. The co-efficient of x^{401} in the expansion of $(1 + x + x^2 + ... + x^9)^{-1}$, $(|x| < 1)$ is -

(A) 1 (B) -1 (C) 2 (D) -2

Q9. The value of $\sum_{r=0}^{s}\sum_{\substack{s=1 \\ r\le s}}^{n} {}^nC_s\,{}^sC_r$ is -

(A) $3^n - 1$ (B) $3^n + 1$

(C) 3^n (D) $3(3^n - 1)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 10 to 11)

The numerically greatest term in the expansion of $(x + a)^n$ is given by $\frac{n+1}{\left|\frac{x}{a}\right|+1} = k$ (say)

(A) If k is an integer then T_k and T_{k+1} are the numerically greatest term.

(B) If k is not an integer. Let m is its integral part then T_{m+1} is the numerically greatest term.

Q10. The numerically greatest term in the expansion of $(1-2x)^8$, when x = 2 is -

(A) ${}^8C_6\,4^6$ (B) ${}^8C_4 4^4$

(C) 2^{17} (D) None of these

Q11. Magnitude wise the greatest term in the expansion of $(3 - 2x)^9$ when x = 1 is -

(A) ${}^9C_2 3^7 2^2$ (B) ${}^9C_3 3^6 2^3$

(C) ${}^9C_4 3^5 2^4$ (D) Both (B) and (C)

Paragraph # 2 (Ques. 12 to 16)

Consider $(1 + x + x^2)^{2n} = \sum_{r=0}^{r=4n} a_r x^r$, where a_0, a_1, a_2,.....a_{4n} are real numbers and n is a positive integer.

Q12. The value of $\sum_{r=0}^{n-1} a_{2r}$ is -

(A) $\frac{9^n - 2a_{2n} - 1}{4}$ (B) $\frac{9^n + 2a_{2n} + 1}{4}$

(C) $\frac{9^n - 2a_{2n} + 1}{4}$ (D) $\frac{9^n + 2a_{2n} - 1}{4}$

Q13. The value of $\sum_{r=1}^{n} a_{2r-1}$ is -

(A) $\left(\frac{9^n - 1}{2}\right)$ (B) $\left(\frac{3^{2n} - 1}{4}\right)$

(C) $\left(\frac{3^{2n} + 1}{4}\right)$ (D) $\left(\frac{9^n + 1}{2}\right)$

Q14. The value of a_2 is -

(A) ${}^{4n+1}C_2$ (B) ${}^{3n+1}C_2$

(C) ${}^{2n+1}C_2$ (D) ${}^{n+1}C_2$

Q15. The value of a_{4n-1} is -

(A) $2n$

(B) $2n^2 + 4n$

(C) $2n + 3$

(D) $2n^2 + 3n$

Q16. The correct statement is -

(A) $a_r = a_{n-r}$, $0 \le r \le n$

(B) $a_{n-r} = a_{n+r}$, $0 \le r \le n$

(C) $a_r = a_{2n-r}$, $0 \le r \le 2n$

(D) $a_r = a_{4n-r}$, $0 \le r \le 4n$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R), (S) and (T). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17.

Column- I	**Column- II**
(A) The sum of the series $\sum_{r=0}^{10} {}^{20}C_r$ is	(P) $-{}^{10}C_5$
(B) The coefficient of x^{53} in $\sum_{r=0}^{100} {}^{100}C_r (x-3)^{100-r} 2^r$ is	(Q) ${}^{100}C_4$
(C) $({}^{10}C_0)^2 - ({}^{10}C_1)^2 + \ldots\ldots\ldots - ({}^{10}C_9)^2 + ({}^{10}C_{10})^2$ equals	(R) $2^{19} + \frac{1}{2}{}^{20}C_{10}$
(D) The value of ${}^{95}C_4 + \sum_{j=1}^{5} {}^{100-j}C_3$ is	(S) $-{}^{100}C_{53}$

Q18.

Column-I	**Column-II**
(A) Number of distinct terms in the expansion of $(x + y - z)^{16}$	(P) 2^{12}
(B) Number of terms in the expansion of $\left(x+\sqrt{x^2-1}\right)^6 + \left(x-\sqrt{x^2-1}\right)^6$	(Q) 97
(C) The number of irrational terms in $\left(\sqrt[8]{5} + \sqrt[6]{2}\right)^{100}$	(R) 4
(D) The sum of numerical coefficients in the expansion of $\left(1 + \frac{x}{3} + \frac{2y}{3}\right)^{12}$	(S) 153

TRIPLE MATCH COLUMN TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column - 1 : It contains binomial expressions.
Column - 2 : It contains number of rational terms.
Column - 3 : It contains number of irrational terms.

Column - 1	Column - 2	Column - 3
(I) $\left(5^{\frac{1}{2}}+7^{\frac{1}{8}}\right)^{1024}$	(i) 251	(P) 896
(II) $\left(5^{\frac{1}{2}}+7^{\frac{1}{8}}\right)^{1000}$	(ii) 501	(Q) 250
(III) $\left(9^{\frac{1}{4}}+8^{\frac{1}{6}}\right)^{500}$	(iii) 126	(R) 875
(IV) $\left(9^{\frac{1}{4}}+8^{\frac{1}{6}}\right)^{1000}$	(iv) 129	(S) 500

Q19. Which of the following options is the only correct combination?
(A) (IV) (iii) (P)
(B) (II) (iv) (S)
(C) (IV) (ii) (S)
(D) (III) (i) (P)

Q20. Which of the following options is the only correct combination?
(A) (III) (i) (S)
(B) (I) (iv) (P)
(C) (I) (iv) (R)
(D) (I) (ii) (S)

Q21. Which of the following options is the only correct combination?
(A) (II) (iv) (Q)
(B) (IV) (iii) (R)
(C) (III) (i) (S)
(D) (III) (i) (Q)

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q22. If $(1+x)^{15}=C_0+C_1.x+C_2.x^2+....+C_{15}.x^{15}$ and $C_2+2C_3+3C_4+....+14C_{15}=a.2^{14}+1$, then find a.

Q23. $\binom{35}{6}+\sum_{r=0}^{10}\binom{45-r}{5}=\binom{x}{y}$,
then x – y is equal to ?

Q24. If ${}^{n+1}C_2+2\,({}^2C_2+{}^3C_2+{}^4C_2+.....+{}^nC_2)=1^2+2^2+3^2+.....+100^2$, then find n.

SUBJECTIVE TYPE

Q25. Prove that : $1^2 . C_0 + 2^2 C_1 + 3^2 . C_2 + 4^2 . C_3 + \ldots\ldots\ldots + (n+1)^2 . C_n = 2^{n-2}(n+1)(n+4)$.

Q26. Prove that :

$$\frac{1}{2}{}^nC_1 - \frac{2}{3}{}^nC_2 + \frac{3}{4}{}^nC_3 - \frac{4}{5}{}^nC_4 + \ldots\ldots + \frac{(-1)^{n+1}n}{n+1} \cdot {}^nC_n = \frac{1}{n+1}$$

Q27. Prove that : $({}^{2n}C_1)^2 + 2.({}^{2n}C_2)^2 + 3.({}^{2n}C_3)^2 + \ldots\ldots + 2n.({}^{2n}C_{2n})^2 = \frac{(4n-1)!}{[(2n-1)!]^2}$

Q28. Prove that :

$$\sum_{i=0}^{r}\binom{n+i}{k} = \binom{n+r+1}{k+1} - \binom{n}{k+1}$$

Q29. If n and k are positive integers, show that :

$$2^k\binom{n}{0}\binom{n}{k} - 2^{k-1}\binom{n}{1}\binom{n-1}{k-1} + 2^{k-2}\binom{n}{2}\binom{n-2}{k-2}\ldots\ldots$$

$$(-1)^k\binom{n}{k}\binom{n-k}{0} = \binom{n}{k}$$

Q30. For any positive integers m, n (with $n \geq m$), let $\binom{n}{m} = {}^nC_m$.

(i) Prove that :

$$\binom{n}{m} + \binom{n-1}{m} + \binom{n-2}{m} + \ldots\ldots\ldots + \binom{m}{m} = \binom{n+1}{m+1}$$

(ii) Prove that : :

$$\binom{n}{m} + 2\binom{n-1}{m} + 3\binom{n-2}{m} + \ldots\ldots\ldots + (n-m+1)\binom{m}{m} = \binom{n+2}{m+2}$$

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
D	A	A	B	C	C	D	D	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	D	A	B	C	C	A	C	A	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	C	B	C	A	B	C	B	C	A
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
D	C	B	A	A	C	B	C	D	C

Ex - 1A

1. B,D **2.** B,C,D **3.** C,D **4.** A,B,C,D
5. A,B,C,D **6.** A,B,C **7.** C **8.** B
9. A,B,C **10.** C,D **11.** B **12.** A **13.** C
14. C **15.** A - R ; B - S ; C - P; D - Q **16.** C **17.** r = 6
18. n = 12 **19.** 40 **20.** (a) $T_3 = \frac{5}{12}$ (b) $T_6 = 7$ **24.** ${}^{30}C_{15}$

25. $(101)^{50} > (100)^{50} + (99)^{50}$ **26.** 3660

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	C	B	C	A	B	B	C	B	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	A	B	C	C	C	A	A	C	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	C	D	D	C	B	A	D	A	D
31	**32**								
A	C								

Ex - 2A

1. B,C,D **2.** A,D **3.** A,D **4.** A,B,C

5. A **6.** A,B **7.** A,B,C,D **8.** B **9.** A

10. C **11.** D **12.** C **13.** B **14.** C

15. A **16.** D **17.** A → R; B → S; C → P; D → Q

18. (A - S, B - R, C -Q, D -P)

19. C **20.** B **21.** D **22.** 13 **23.** 6 or 40

24. n = 100

-: Important Notes & Formulas :-